SUPRIYA SARKER

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EDUCATION

University of Arkansas, Fayetteville — *January 2025 - May 2028 (Expected)*
Ph.D. in Industrial Engineering

University of Tennessee, Knoxville — *August 2023 - December 2024*
MS in Computer Engineering

Chittagong University of Engineering & Technology — *June 2016 - February 2022*
MS in Computer Science & Engineering

Chittagong University of Engineering & Technology — *March 2010 - December 2014*
BS in Computer Science & Engineering

HONORS AND AWARDS

P.E.O. Scholar Award Nominee — *P.E.O. Sisterhood, 2027–2028*
P.K. Kuroda Endowed Graduate Fellowship in Engineering — *University of Arkansas, 2026–2027*
Best Paper Award – Honorable Mention — *ICAICT, 2020*

SOFTWARE AND TOOLS

Programming Languages and Tools	Python, SQL, C, Java, MATLAB, HTML, LaTeX
Development Tools	Git/GitHub, Jupyter Notebook, Google Colab, Excel
AI/LLM Frameworks	PyTorch, TensorFlow, Hugging Face, LangChain, Vector Databases, Prompt Engineering, LLM API Integration
Data Science Libraries	Pandas, NumPy, SciPy, Matplotlib, Scikit-learn

RESEARCH METHODOLOGIES

Machine Learning, Deep Learning, Reinforcement Learning, Generative AI, Multimodal AI, Agentic AI Systems, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

RESEARCH EXPERIENCES

University of Arkansas, Fayetteville — August 2026 - Present
Graduate Research Assistant

- Developing a context-aware LLM framework using RAG and plant-specific risk data to prioritize maintenance work orders in nuclear power plants.

University of Tennessee, Knoxville — January 2026 - July 2026
Graduate Research Assistant

- Designed an agentic AI system leveraging LLMs to support autonomous inspection and quality assessment in manufacturing, integrating multi-step reasoning with multimodal data and domain-specific rules to detect, interpret, and classify metal deposition anomalies. **GitHub**

Oak Ridge National Laboratory — August 2025 - December 2025
Graduate Research Intern

- Designed privacy-preserving hyperparameter optimization for federated foundation models using adaptive trial selection and privacy budget allocation.

University of Tennessee, Knoxville — August 2023 - December 2024
Graduate Teaching and Research Assistant

- Built the Beacon real-world blackout traffic dataset and applied Rainbow DQN to control robot vehicles in reconstructed traffic, reducing wait times by up to 82.6%. **GitHub**

Chittagong University of Engineering & Technology — March 2017 - February 2022
Graduate Research Assistant

- Built an LSTM classifier for seven driving maneuvers using sensor dataset, achieving over 98% per-class accuracy. **GitHub**